



By Nitesh Singh

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SOC

Analyst L1 For Free

Defronix Cyber Security → <https://defronix.com>

Windows Event Logs



Windows event logs are forensic timeline of everything happening inside Windows.

Every action generates telemetry:

User login → Event generated
Process starts → Event generated
Service starts → Event generated
Policy change → Event generated

Where logs are stored

C:\Windows\System32\winevt\Logs\

File format: .evtx

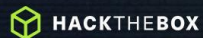
Why they are CRITICAL for SOC

Because logs answer:

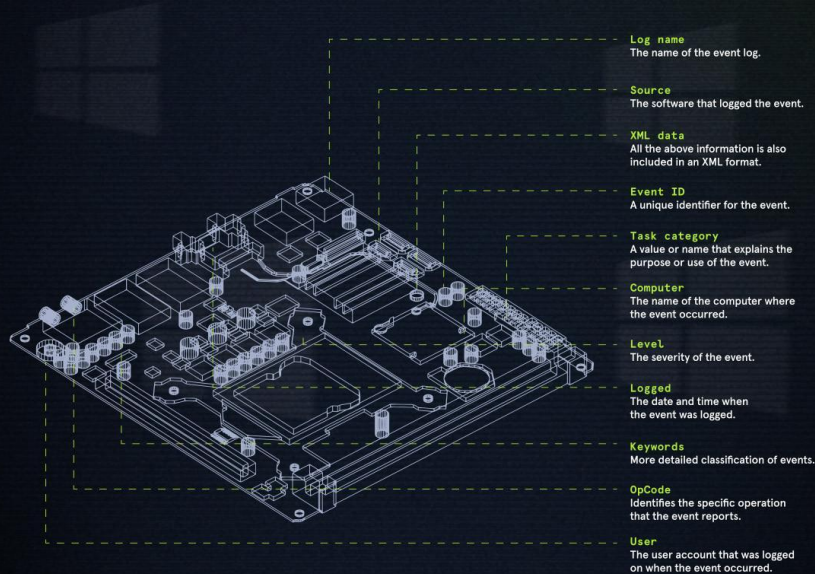
WHO did WHAT, WHEN, FROM
WHERE, and HOW

Without logs → no investigation.

WINDOWS LOGGING ARCHITECTURE

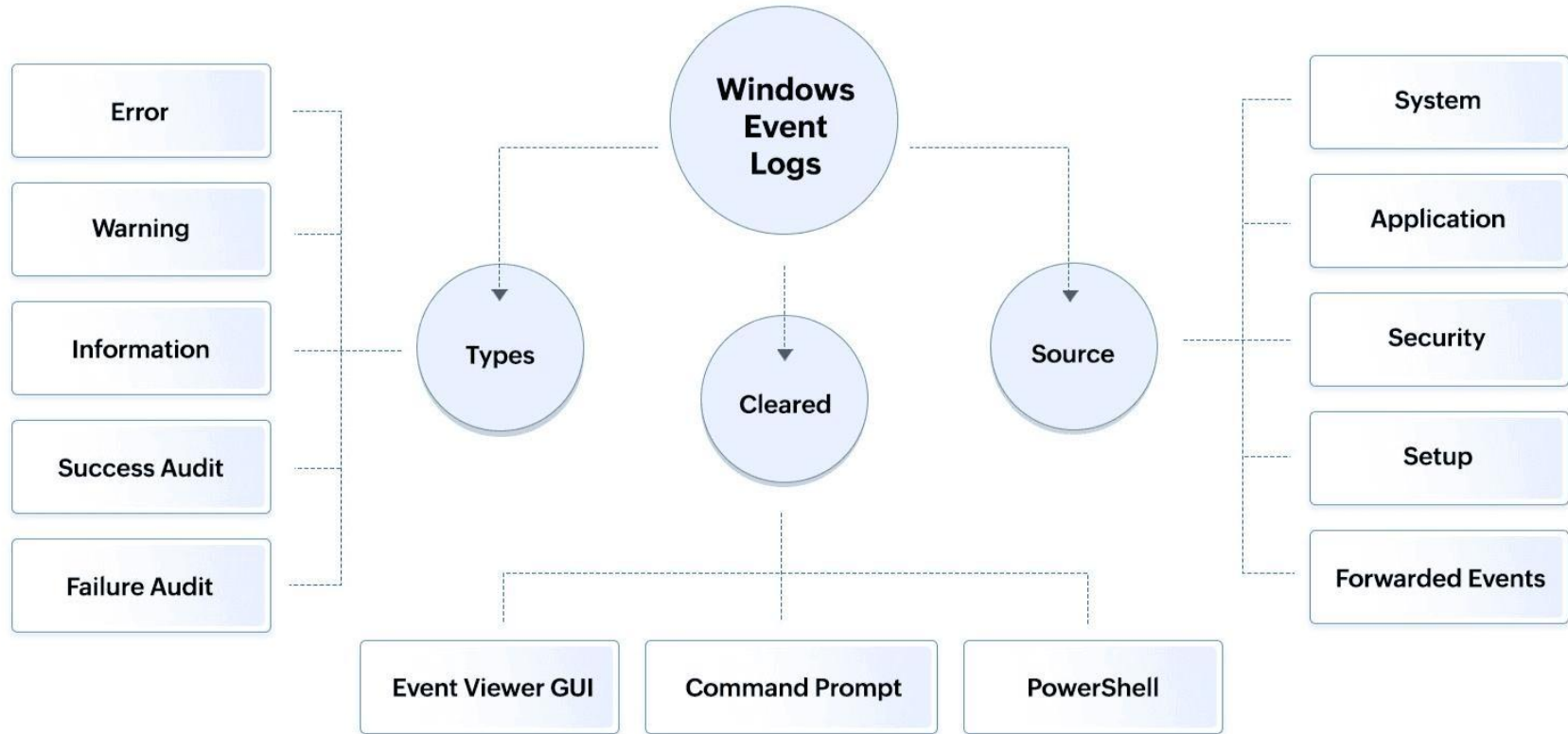


The anatomy of a Windows event log



Source : hackthebox

WINDOWS LOGGING ARCHITECTURE





Step-by-step flow

1. Action happens (login, process, etc.)
2. Event Provider generates event
3. Windows Event Log Service collects it
4. Event written to EVTX file
5. Viewed in Event Viewer / SIEM

Key Components

Event Provider

- Generates logs
- Example:
 - Security provider
 - PowerShell

Event Channel (Log Type)

- Security
- System
- Application

Event Log Service

- Stores logs



LOGON TYPES (VERY IMPORTANT)

What are Security Logs?

Security logs track:

- Authentication
- Authorization
- Privileges
- Account changes

These are **your primary investigation logs**

Type	Meaning	SOC Use
2	Interactive	Local login
3	Network	SMB access
10	Remote	RDP
7	Unlock	Session reuse

SOC must map:

Logon Type + IP + User = Attack pattern

SECURITY EVENT LOGS – KEY EVENT IDs

IMPORTANT SECURITY EVENT IDs IN WINDOWS SECURITY LOGS



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1

LOGON & AUTHENTICATION EVENTS

Track user logon activities and detect unauthorized access.

4624

Successful Logon

Monitor logon. (2=Interactive, 10=RDP, 3=Network)

4625

Failed Logon

Detect brute-force / credential stuffing attacks.

4634

Logoff

User logoff event.

4647

User Initiated Logoff

User explicitly logged off.

4648

Logon using Explicit Credentials

Possible credential abuse (RunAs, net use, scheduled tasks).

4672

Special Privileges Assigned

Admin / privileged logon (Very Important).

2

ACCOUNT MANAGEMENT EVENTS

Monitor user account lifecycle changes.

4720

User Account Created

New user account created.

4722

Account Enabled

User account enabled.

4723

Password Change Attempt

User changed own password.

4724

Password Reset (by Admin)

Password reset by administrator.

4725

Account Disabled

User account disabled.

4726

Account Deleted

User account deleted.

3

GROUP & PRIVILEGE MANAGEMENT

Detect privilege escalation and group membership changes.

4728

Member Added to Global Group

User added to global group.

4729

Member Removed from Global Group

User removed from global group.

4732

Member Added to Local Group

User added to local security group.

4733

Member Removed from Local Group

User removed from local group.

4756

Member Added to Universal Group

User added to universal group.

4757

Member Removed from Universal Group

User removed from universal group.

CRITICAL FOCUS: Adding user to Administrators / Domain Admins = **HIGH ALERT**

4

PROCESS CREATION & EXECUTION

Detect malicious processes and code execution.

4688

Process Created

Most Important Event!
Track new process, parent process, command line, user.

4689

Process Terminated

Process termination event.

5

PRIVILEGE USE EVENTS

Monitor use of sensitive privileges and operations.

4673

Sensitive Privilege Used

User exercised a sensitive privilege (e.g., SeDebugPrivilege).

4674

Operation on Privileged Object

An operation was performed on a privileged object.

6

AUTHENTICATION (KERBEROS / NTLM)

Monitor domain authentication and detect attacks.

4768

Kerberos TGT Request

Initial ticket granting ticket requested.

4769

Kerberos Service Ticket

Service ticket requested.

4771

Kerberos Pre-auth Failed

Kerberos pre-authentication failed.

4776

NTLM Authentication

NTLM authentication attempt.

DETECTION USE: Kerberos failures = password spraying / brute force. NTLM usage = legacy auth / potential abuse.

7

OBJECT ACCESS EVENTS

Track access to files, registry and other objects.

4656

Handle Requested

A handle was requested to an object.

4663

Object Access Attempt

An attempt was made to access an object.

4658

Handle Closed

A handle to an object was closed.

9

LOG TAMPERING (CRITICAL ALERT)

Detect attempts to clear or tamper with logs.

1102

Security Log Cleared

IMMEDIATE INCIDENT RESPONSE REQUIRED

10

SYSTEM & SECURITY STATE EVENTS

Track system level security events.

4608

Windows Started

System startup.

4609

Windows Shutdown

System shutdown.

4616

System Time Changed

System time was changed.

DETECTION: Time change can indicate log manipulation.

8

POLICY & AUDIT CHANGES

Detect changes to policies and auditing.

4719

Audit Policy Changed

Audit policy was changed.

4739

Domain Policy Changed

Domain policy was changed.

HIGH RISK: Attackers may disable auditing to hide their activities.

ATTACK DETECTION CHEAT SHEET

BRUTE FORCE ATTACK
Multiple 4625 Failed Logon events from same IP/user.

SUCCESSFUL ATTACK
4625 followed by 4624 Successful Logon.

PRIVILEGE ESCALATION
4672 + Group membership change events.

MALWARE EXECUTION
4688 with suspicious process or command line.

PERSISTENCE
New account (4720) or service creation (7045) or scheduled task.

LOG TAMPERING
1102 Security Log Cleared.

TIPS FOR ANALYSTS

Always check Logon Type in 4624 events.

Correlate multiple events for accurate detection.

Investigate unusual logon times, locations and privileges.

Enable Advanced Audit Policy for comprehensive logging.

Regularly review and monitor critical Event IDs.

ALERT. ANALYZE. RESPOND. STAY SECURE.



What are System Logs?

Tracks:

- OS behavior
- Drivers
- Hardware
- Boot events

Important Events

6005 — System started

6006 — System shutdown

6008 — Unexpected shutdown

7045 — Service installed 🔥

SOC USE

- Detect system crashes (possible attack)
- Detect malicious service installation



WINDOWS SYSTEM LOGS – KEY EVENT IDs

IMPORTANT SYSTEM LOG EVENTS FOR TROUBLESHOOTING & SYSTEM HEALTH MONITORING

1



SYSTEM STARTUP & SHUTDOWN EVENTS

Track system boot, shutdown and uptime.

EVENT ID	DESCRIPTION	
6005	Event Log Service Started	System boot / startup.
6006	Event Log Service Stopped	Normal shutdown.
6008	Unexpected Shutdown	System crashed or lost power unexpectedly.
6013	System Uptime	Shows how long the system has been running.

2



POWER & REBOOT EVENTS

Identify power issues and reboot reasons.

EVENT ID	DESCRIPTION	
41	Kernel-Power	System rebooted without clean shutdown (power failure, crash, BSOD).
1074	Planned Shutdown/Restart	System shutdown or restart initiated by user or process.
109	Kernel Power Critical Error	Critical power-related error.

3



DRIVER & HARDWARE EVENTS

Hardware and driver related issues.

EVENT ID	DESCRIPTION	
219	Driver Failed to Load	A driver failed to load during system start.
225	Device Not Migrated	Device not migrated properly.
11	Disk Controller Error	Disk controller detected an error.
15	Device Not Ready	The device is not ready for use.
7	Bad Block Detected	A bad block was detected on the disk.

4



DISK & STORAGE EVENTS

Monitor disk health and file system issues.

EVENT ID	DESCRIPTION	
51	Disk Warning	An I/O operation was retried or failed on the disk.
55	NTFS Corruption	The file system structure is corrupted.
98	NTFS Volume Corruption	A corruption was detected on the volume.

5



SERVICE CONTROL MANAGER EVENTS

Monitor Windows services and failures.

EVENT ID	DESCRIPTION	
7000	Service Failed to Start	The service failed to start.
7001	Service Dependency Failed	The service depends on another service that failed to start.
7009	Service Timeout	The service did not start in the expected time.
7011	Service Timeout (No Response)	Timeout waiting for service to respond.
7034	Service Crashed Unexpectedly	A running service crashed.
7036	Service State Changed	Service state changed (Running/Stopped/Paused).

6



NETWORK & CONNECTIVITY EVENTS

Network adapter and connectivity issues.

EVENT ID	DESCRIPTION	
4201	Network Adapter Connected	A network adapter is connected to the system.
4202	Network Adapter Disconnected	A network adapter is disconnected from the system.
1014	DNS Client Events	DNS name resolution has failed.

7



TIME & SYSTEM INTEGRITY EVENTS

Time synchronization and system integrity.

EVENT ID	DESCRIPTION	
1	Time Synchronized	The system time was synchronized.
36	Time Service Error	Time service encountered an error.

8



CRASH & ERROR REPORTING EVENTS

Track system crashes and errors.

EVENT ID	DESCRIPTION	
1001	BugCheck (BSOD)	A Blue Screen of Death occurred.
1000	Application Error	An application crashed or stopped working.

9



CRITICAL SYSTEM HEALTH INDICATORS

High priority events to watch.

1

41

Unexpected reboot or power issue

1

6008

Unexpected shutdown

1

7, 11, 51, 55, 98

Disk / file system problems

1

7034

Service crashed unexpectedly

1

1001

Blue Screen of Death (BSOD)



SYSTEM LOGS CHEAT SHEET



System Crash / Reboot
41 + 6008 + 1001



Disk Failure
7 / 51 / 55 / 98



Service Failure
7000 / 7034



Driver Issue
219 / 225



Network Issue
1014 (DNS)

★ **PRO TIP:** Use System Logs with Security Logs together for complete visibility of issues and attacks.

COLLECT. CORRELATE. INVESTIGATE. RESOLVE.



What they track

- App crashes
- Errors
- Software activity

SOC Relevance

- Detect exploit attempts
- Identify malicious app behavior



WINDOWS APPLICATION LOGS – KEY EVENT IDs



IMPORTANT APPLICATION LOG EVENTS FOR MONITORING & TROUBLESHOOTING



1 APPLICATION START & GENERAL EVENTS

Identify application start, stop and general information.

EVENT ID	DESCRIPTION	
1000	Application Error	An application crashed or stopped working unexpectedly.
1001	Application Hang	The application is not responding.
1002	Application Unresponsive	The application is unresponsive for a period of time.
1003	Application Restart	The application was restarted.
1004	Application Start	The application started successfully.
1005	Application Stop	The application stopped successfully.

2 APPLICATION CRASH & FAILURE EVENTS

Detect critical application failures and crashes.

EVENT ID	DESCRIPTION	
1006	Application Crash	The application crashed.
1007	Application Fault	A fault occurred in the application.
1026	.NET Runtime Error	.NET Framework application error.
1027	.NET Runtime Warning	.NET Framework application warning.
1028	.NET Runtime Information	.NET Framework informational event.
1029	.NET Runtime Debug	.NET Framework debugging event.

3 APPLICATION INSTALL & UPDATE EVENTS

Track installation, configuration and updates.

EVENT ID	DESCRIPTION	
1030	Application Install	The application was installed.
1031	Application Uninstall	The application was uninstalled.
1032	Application Update	The application was updated.
1033	Configuration Change	Application configuration was changed.
1034	Configuration Error	Error in application configuration.

4 SECURITY & PERMISSION EVENTS

Monitor security and access related issues.

EVENT ID	DESCRIPTION	
1040	Application Access Denied	Access to the application was denied.
1041	Invalid Credentials	Invalid credentials used by the application.
1042	Login Failure	Application login failed.
1043	Permission Change	Permissions for the application were changed.

5 PERFORMANCE & RESOURCE EVENTS

Track performance and resource issues.

EVENT ID	DESCRIPTION	
1050	Performance Warning	Application performance degraded.
1051	Performance Error	Application performance critical error.
1052	Resource Warning	Insufficient resources available.
1053	Resource Error	Critical resource

6 APPLICATION & SYSTEM INTEGRATION EVENTS

Track communication and integration events.

EVENT ID	DESCRIPTION	
1060	Service Communication Failure	Application failed to communicate with service.
1061	External System Error	Error communicating with external system.
1062	Message Queue Error	Application message queue error.
1063	Email Service Error	Application email service error.

Example:

Application: Word crashed

Could be exploit



Definition

A **process** is:

A running instance of a program with its own:

- Memory space
- Threads
- Resources
- Security context

Simple View

Program (on disk) → executed → becomes a Process (in memory)

Example:

chrome.exe (file) → running → process with PID



Key Components of a Process

1. PID (Process ID)

- Unique identifier

Example:

PID: 4321

2. Virtual Memory

- Each process gets isolated memory

SOC:

- Used in process injection

3. Threads

- Units of execution inside process

One process = multiple threads

4. Handles

- References to system resources
- Files, registry, network

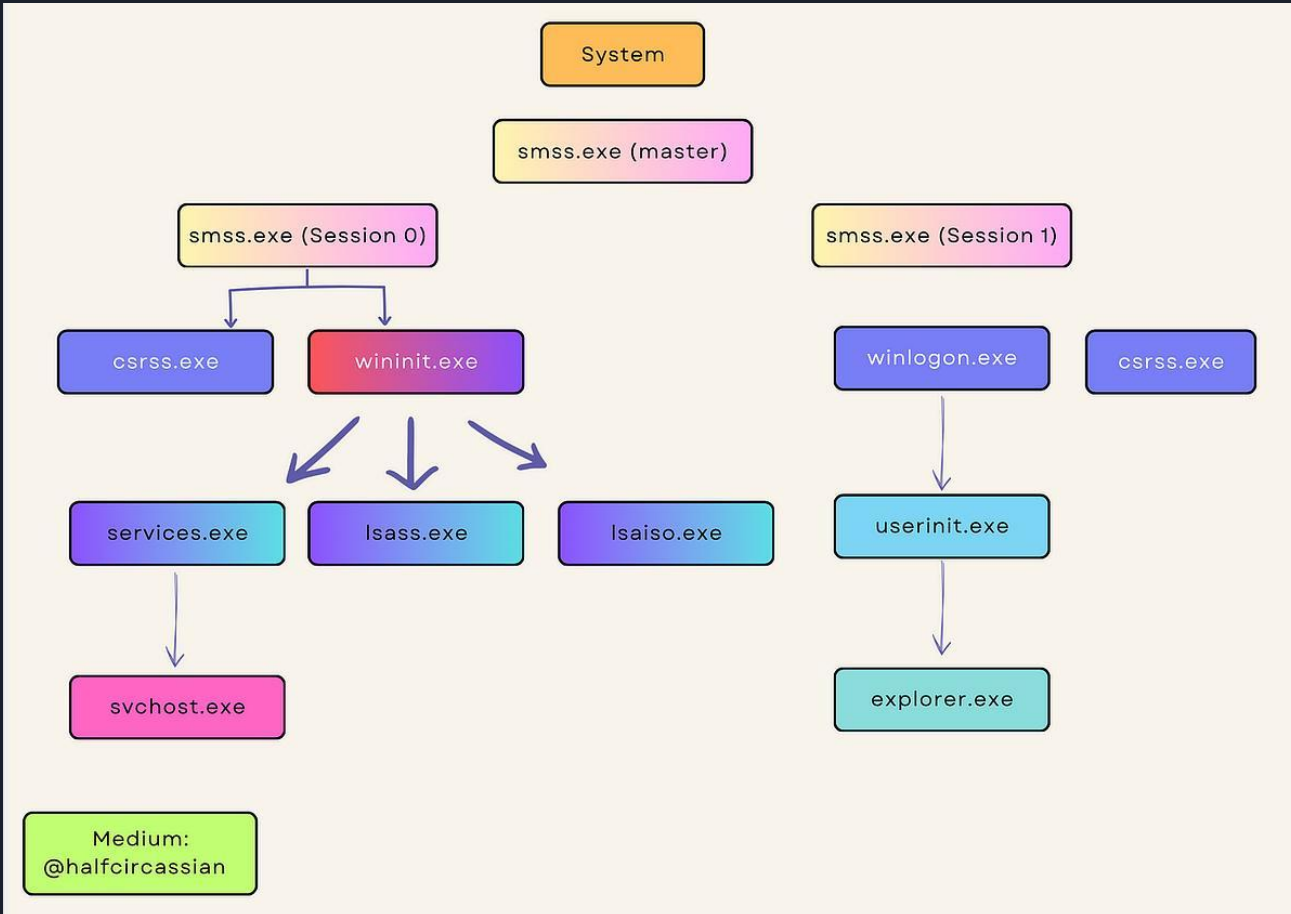
5. Security Context

- Runs as a user

Example:

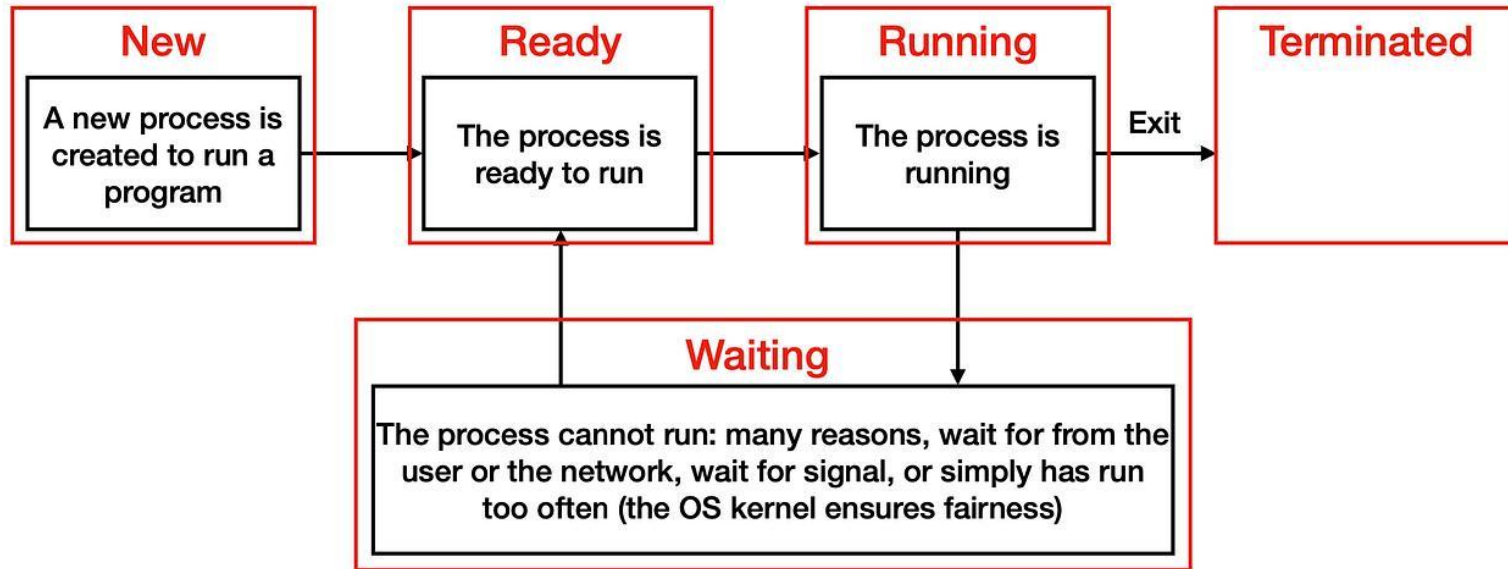
User: SYSTEM / Admin / User

Windows Processes Architecture





Process Life Cycle





Step-by-step

1. User/system triggers execution
2. Win32 API called
3. Kernel creates process object
4. Memory allocated
5. Threads created
6. Process starts execution

Parent–Child Relationship (VERY IMPORTANT)

Every process is created by another process.

Example

explorer.exe → chrome.exe

SOC Insight

If parent-child chain is abnormal → suspicious

Example:

winword.exe → powershell.exe ❌



CRITICAL WINDOWS SYSTEM PROCESSES



ESSENTIAL PROCESSES THAT KEEP WINDOWS RUNNING SMOOTHLY



PROCESS NAME	PROCESS FILE	PID RANGE (TYPICAL)	DESCRIPTION	IMPORTANCE	IMPACT IF TERMINATED
System	ntoskrnl.exe	4 – 500	The Windows NT OS kernel. Manages hardware, memory, processes and system resources.	CRITICAL	⚠ System will crash immediately (BSOD).
smss.exe	smss.exe	500 – 600	Session Manager Subsystem. Starts user sessions and critical system processes.	CRITICAL	⚠ System will fail to start or shut down.
csrss.exe	csrss.exe	600 – 900	Client/Server Runtime Subsystem. Handles Win32 subsystem and console windows.	CRITICAL	⚠ Applications and system UI will stop working.
wininit.exe	wininit.exe	900 – 1000	Windows initialization process. Starts services and userinit.exe.	CRITICAL	⚠ System may become unusable.
services.exe	services.exe	1000 – 1500	Service Control Manager. Manages Windows services.	CRITICAL	⚠ Many services will stop, system instability.
lsass.exe	lsass.exe	600 – 1200	Local Security Authority Subsystem. Handles authentication, security policy and user login.	CRITICAL	⚠ Logon will fail, security functions stop.
svchost.exe	svchost.exe	1000 – 6000+	Generic host process for Windows services. Multiple instances run critical services.	CRITICAL	⚠ Dependent services will stop.
explorer.exe	explorer.exe	2000 – 4000	Windows Explorer. Manages desktop, taskbar, Start menu and file explorer.	HIGH	⚠ Desktop, taskbar and UI will disappear.
spoolsv.exe	spoolsv.exe	1500 – 3000	Print Spooler. Manages print jobs and printer communication.	HIGH	⚠ Printing will not work.
taskhostw.exe	taskhostw.exe	2000 – 5000	Hosts tasks scheduled by user or system (Task Scheduler).	MEDIUM	⚠ Scheduled tasks may not run.
dwm.exe	dwm.exe	1500 – 3000	Desktop Window Manager. Provides visual effects and window management.	MEDIUM	⚠ Affects UI performance and visuals.
svchost.exe (DNS Client)	svchost.exe	1000 – 6000+	Hosts DNS Client service. Resolves domain names.	MEDIUM	⚠ Internet name resolution may fail.
svchost.exe (Power)	svchost.exe	1000 – 6000+	Hosts Power services. Manages power settings.	MEDIUM	⚠ Power management issues.
wuauclt.exe	wuauclt.exe	2000 – 4000	Windows Update Client. Checks for and installs updates.	LOW	ℹ Updates will not work.
MsMpEng.exe	MsMpEng.exe	2000 – 5000	Microsoft Defender Antivirus engine process.	LOW	ℹ System protection will be disabled.

⚠ WARNING – DO NOT END THESE PROCESSES

🔍 HOW TO IDENTIFY CRITICAL PROCESSES?

⚙ BEST PRACTICES

Critical Windows System Processes

Defronix Cyber Security → <https://defronix.com>

Windows Processes Tree



Standard Process Tree



SOC USE

Any deviation = investigation

Example:

winword.exe → cmd.exe → powershell.exe





PROCESS TYPES

1. User Processes

- Applications

2. System Processes

- OS-level

3. Background Processes

- Services



1. Process Injection

Concept

Inject code into another process

Example

malware → inject → explorer.exe

SOC Indicator

- Suspicious child processes
- Memory anomalies

2. DLL Injection

- Inject malicious DLL

3. Process Hollowing

Concept

- Start legit process
- Replace its memory

Example:

notepad.exe → replaced with malware



4. LOLBins (Living Off the Land)

Using legit tools:

- powershell.exe
- cmd.exe
- rundll32.exe

SOC Indicator

powershell.exe with encoded command



SOC Steps

1. Identify process (4688)
2. Check parent process
3. Check command line
4. Check user context
5. Check network activity
6. Correlate with logs

COMMON RED FLAGS

Suspicious patterns:

- Word → PowerShell
- Browser → cmd.exe
- svchost.exe spawning child process
- Random executable from temp folder

SOC MINDSET

When you see a process:

Don't ask:

✗ "What is this process?"

Ask:

Why is this process running?

Who started it?

What did it do next?



Definition

A **Windows Service** is:

- A program that runs in the **background**.
- Without user interaction.
- Often starts **automatically at system boot**.

Simple Understanding

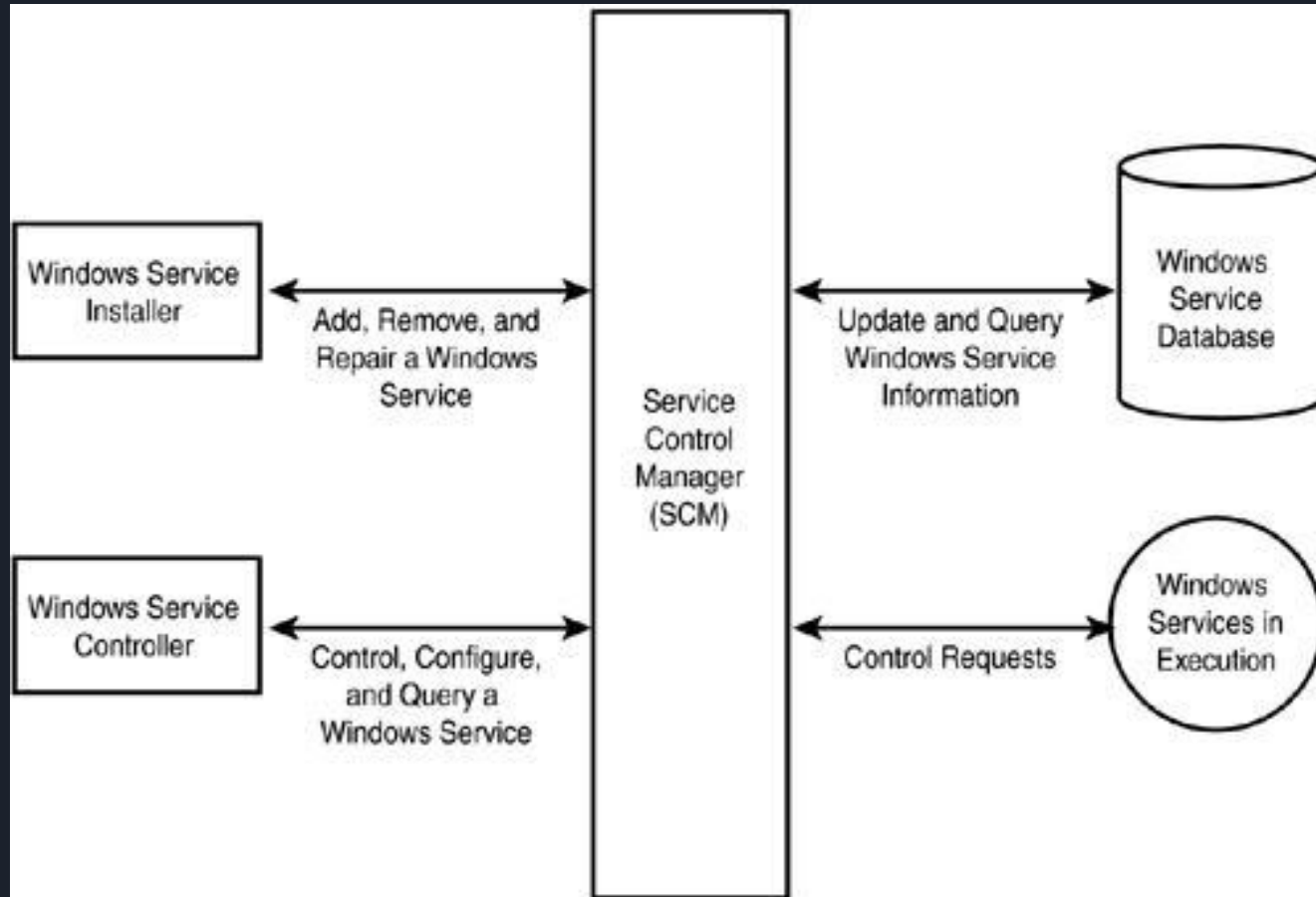
Application → needs user → visible

Service → runs silently → background

Examples of Legit Services

- Windows Update
- Antivirus
- Print Spooler
- DHCP Client
- Remote Desktop Service

HOW SERVICES WORK INTERNALLY





Core Component: SCM (Service Control Manager)

What is SCM?

Central controller of all services

Responsibilities:

- Start services
- Stop services
- Manage service state
- Maintain service database

Service Execution Flow

System Boot



SCM starts



SCM reads registry



Loads services



Services start running



Where are services stored?

In **Registry**:

HKLM\SYSTEM\CurrentControlSet\Services

Each service contains:

- Service Name
- Image Path (executable path)
- Start type
- Permissions

SERVICE TYPES

1. Kernel Services

- Drivers

2. File System Services

- Disk operations

3. User-mode Services

- Normal background services



SERVICE STARTUP TYPES

Type	Meaning
Automatic	Starts at boot
Manual	Starts when needed
Disabled	Cannot start

svchost.exe (VERY IMPORTANT)

What is svchost.exe?

A **container process** that runs multiple services

Example:

svchost.exe → hosts multiple services

SOC Importance

- Malware hides inside svchost
- Hard to detect

HOW SERVICES WORK INTERNALLY



SERVICES VS PROCESSES

Feature	Process	Service
User interaction	Yes	No
Runs	On demand	Background
Persistence	No	Yes



1. Malicious Service Installation

Example:

Service Name: UpdateService

Path: C:\Users\Temp\malware.exe

Looks legit but is malware

2. Service Hijacking

Replace path of existing service:

legit.exe → replaced with malware.exe

3. Privilege Escalation via Services

If service runs as SYSTEM:

attacker gets admin access

4. Persistence

Malware installs service → runs at every reboot



Logs

4624 → login

4688 → powershell.exe

7045 → new service installed

SOC Thinking

Initial access → execution → persistence



SOC Steps

1. Identify service (7045)
2. Check path
3. Check user privileges
4. Check parent process
5. Check related logs
6. Correlate timeline

RED FLAGS (SOC)

Suspicious:

- New service created suddenly
- Service from temp folder
- Random service name
- Service created after PowerShell

SOC MINDSET

When you see a service:

Don't think:

✗ "This is just a background process"

Think:

Is this persistence?

Is this privilege escalation?

Is this attacker foothold?